

Supplementary Material for “DEFT: Differentiable Automatic Test Pattern Generation”

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This document contains the appendices of the paper “DEFT: Differentiable Automatic Test Pattern Generation,” published in the Proceedings of the IEEE International Test Conference (ITC) 2026. Theorem, equation, and figure numbers that are not defined here (e.g., Theorem IV.1, equation DEFT-P1, and Fig. 12) refer to the main paper.

APPENDIX A PROOFS

A. Proof of Theorem IV.1

Proof. Let $\mathcal{X}_D \subseteq \{0, 1\}^n$ be the set of all detecting patterns, i.e., $\mathcal{X}_D = \{x \in \{0, 1\}^n \mid I_f(x) = 1\}$.

a) *P1 solvable $\implies$ P2 optimal value is 1.* Assume equation DEFT-P1 is solvable. Then the set of detecting patterns $\mathcal{X}_D$ is non-empty. Let $x^* \in \mathcal{X}_D$ be one detecting pattern. Define $\theta^* \in [0, 1]^n$ as the deterministic parameter vector corresponding to x^* :

$$\theta_i^* = \begin{cases} 1, & \text{if } x_i^* = 1, \\ 0, & \text{if } x_i^* = 0. \end{cases}$$

For this θ^* , the product distribution $p(x|\theta^*)$ is a Dirac distribution centered at x^* . Therefore,

$$\begin{aligned} J(\theta^*) &= \mathbb{E}_{x \sim p(\cdot|\theta^*)}[I_f(x)] \\ &= \sum_{x \in \{0, 1\}^n} I_f(x) \cdot p(x|\theta^*) \\ &= I_f(x^*) \cdot p(x^*|\theta^*) = 1. \end{aligned}$$

Since $J_f(\theta)$ is the expectation of a binary indicator, it cannot exceed 1. Hence $\max_{\theta \in [0, 1]^n} J_f(\theta) = 1$.

b) *P2 optimal value is 1 $\implies$ P1 solvable.* Assume $\max_{\theta \in [0, 1]^n} J_f(\theta) = 1$. Then there exists some θ^* such that $J(\theta^*) = 1$. By definition,

$$\begin{aligned} J(\theta^*) &= \mathbb{E}_{x \sim p(\cdot|\theta^*)}[I_f(x)] \\ &= \sum_{x \in \{0, 1\}^n} I_f(x) \cdot p(x|\theta^*) \\ &= \sum_{x \in \mathcal{X}_D} p(x|\theta^*). \end{aligned}$$

Thus $J(\theta^*)$ is the total probability mass assigned to detecting patterns. If this mass is 1, then $\mathcal{X}_D$ cannot be empty. Therefore, at least one detecting pattern exists, and equation DEFT-P1 is solvable. $\square$

B. Proof of the Logistic Reparameterization

Lemma A.1 (Logistic Reparameterization). *Let $\hat{\theta} \in \mathbb{R}^n$ be a vector of logits. Let $\epsilon \in \mathbb{R}^n$ be a noise vector with independent components $\epsilon_i \sim \text{Logistic}(0, 1)$, drawn by sampling $u_i \sim \mathcal{U}(0, 1)$ and setting $\epsilon_i = \log(u_i) - \log(1 - u_i)$. Define*

$$\tilde{x}_{hard}(\hat{\theta}, \epsilon)_i = \mathbb{I}(\hat{\theta}_i + \epsilon_i > 0). \quad (1)$$

Then $P(\tilde{x}_{hard}(\hat{\theta}, \epsilon)_i = 1) = \sigma(\hat{\theta}_i)$.

Proof. By definition,

$$\begin{aligned} P((\tilde{x}_{hard})_i = 1) &= P(\hat{\theta}_i + \epsilon_i > 0) \\ &= P(\epsilon_i > -\hat{\theta}_i). \end{aligned}$$

The standard Logistic distribution has cumulative distribution function $F(y) = P(\epsilon_i \leq y) = \frac{1}{1 + e^{-y}} = \sigma(y)$. Therefore,

$$\begin{aligned} P(\epsilon_i > -\hat{\theta}_i) &= 1 - F(-\hat{\theta}_i) \\ &= 1 - \sigma(-\hat{\theta}_i) \\ &= \frac{e^{\hat{\theta}_i}}{1 + e^{\hat{\theta}_i}} = \frac{1}{1 + e^{-\hat{\theta}_i}} = \sigma(\hat{\theta}_i). \end{aligned}$$

$\square$

APPENDIX B

DEFT HYPERPARAMETERS

TABLE I: Key hyperparameter settings used in all DEFT experiments.

Hyperparameter	Setting and Description
Gumbel temperature	τ is annealed from $\tau_{\text{start}} = 3.0$ to $\tau_{\text{end}} = 0.5$ using a cosine schedule. A high initial temperature encourages exploration in the relaxed space, and the low final temperature sharpens the distribution for discrete decisions.
Explore weight	w_{explore} is annealed linearly from $w_{\text{start}} = 1.0$ to $w_{\text{end}} = 0$ over the entire run. This provides strong early-stage guidance from continuous surrogates while ensuring that the final optimization is governed by the exploit objective.
Learning rate and optimizer	A constant learning rate of 0.1 is used for the AdamW optimizer. This value balances training stability and convergence speed across all benchmarks.
Monte Carlo samples	Each batch entry draws $K_{\text{samples}} = 3$ Gumbel perturbations. Final evaluation uses $K_{\text{eval}} = 3 \times \text{batch_size}$ samples. In our experiments, K_{samples} has no significant impact on result quality.
Batch size	Let B_{max} denote the maximum batch size given the available memory and computational resources. We set the batch size as $B_{\text{max}}/4$. We note that number above $B_{\text{max}}/4$ will linearly increase the runtime without significant improvement in result quality.
Epochs	Each DEFT run executes for 1000 epochs. This was sufficient for all faults to converge under the annealing schedule.
Multi-fault and multi-pattern	DEFT optimizes over the entire HTD fault set in each run. For MAC, $T = 1$. For others, $T = 40$. We note that the convergence speed is strongly affected by the solution space, determined by the circuit scale, the number of faults, and the pattern budget. We leave the exploration of more advanced multi-fault and multi-pattern optimization strategies to future work.

APPENDIX C

COMMERCIAL ATPG SCRIPTS

The command snippets below summarize the key commercial-tool scripts used in the experimental flow.¹

S1 Default run for HTD identification

```
add_faults -all
create_patterns
```

S2 Pattern-limited run for HTD identification

```
add_faults -all
create_patterns -Pattern_count
${tp_limit}
```

S3 Commercial baseline on the HTD set

```
read_faults ${HTD}
set_abort_limit 5002
create_patterns -COverage_effort high
-COMPaction_effort Maximum
```

¹For simplicity, prerequisite setup commands such as `read_verilog` are omitted.

²We note that increasing abort limit does not show an obvious pattern count decrease. See Fig. 12.

S4 Commercial finishing step after DEFT

```
read_faults ${final_remaining_faults}
set_abort_limit 500
create_patterns -COverage_effort high
-COMPaction_effort Maximum
```